RAG with Sufficient Context: Selective Generation for Safer Open-Book QA

Aleksandr Gavkovskii Ilya Maksimov Karim Zakirov

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Abstract

We study selective generation in Retrieval-Augmented QA: deciding whether to answer or abstain given a question and retrieved passages. Following the ICLR 2025 *sufficient-context* line of work, we train a gate on the pair $(P(\text{Correct}), \text{Sufficient})$, where the sufficiency label is produced by an LLM autorater on the retrieved passages. We build an end-to-end pipeline (retriever $\rightarrow$ generator $\rightarrow$ autorater $\rightarrow$ gate $\rightarrow$ split-conformal calibrator) and run it end-to-end on 500 HotPotQA validation examples with two generator configurations: (i) a **Mistral-7B-Instruct-v0.3** run with inline self-reported confidence, and (ii) a **Qwen2.5-3B-Instruct** run with 5-sample self-reported confidence. On the Qwen run the combined gate reduces out-of-fold AURC from **0.256** (confidence only) to **0.172** (proposed), a **33% relative reduction** that is statistically significant under a paired bootstrap ($p \approx 0.000$, 500 resamples). Expected Calibration Error drops from 0.603 to 0.044 and AUROC rises from 0.498 to 0.697, showing that the sufficiency signal restores a usable confidence ordering that Qwen's self-reported confidence alone does not provide. Split-conformal prediction at target risk $\alpha = 0.1$ is the one place the pipeline under-delivers – with only 94 calibration points the realised risk is 0.286 at 6.4% coverage, which we attribute to small-n conformal quantile noise.

1 Motivation

Open-book QA systems that stack a retriever on top of an instruction-tuned LLM are an appealing way to ground answers in documents. In practice they still hallucinate: the model answers confidently even when the retrieved passages do not contain the answer. Confidence alone is not enough to decide whether to answer – *a confident model with insufficient context is exactly the failure mode we want to catch.*

Following the ICLR 2025 work on *sufficient context* [1], we train a selective-generation gate that combines two signals:

1. **Model confidence** that the produced answer is correct, $P(\text{Correct})$.
2. **Context sufficiency**, an LLM-as-a-judge decision on whether the retrieved passages actually support a definitive answer.

The hypothesis is that these signals are *complementary*: sufficiency flags a failure mode that self-reported confidence misses, so a gate trained on the pair should Pareto-dominate the confidence-only baseline on the risk–coverage curve.

2 Method

2.1 Pipeline overview

```
Question + Passages
        |
        v
   +-----------+ +-------------+
   | Generator | ----> | Confidence |
   +-----------+ +------+------+
        | |
        v v
   +-----------+ +-------------+
   | Autorater | ----> | Gate |
   | Sufficient| | g(c, s, ...)|
   +-----------+ +------+------+
                          v
                   Answer / Abstain
```

The gate is trained only on answerable outputs (correct vs. hallucinate). We compute 5-fold cross-validated OOF scores to avoid look-ahead bias when plotting selective curves, and we apply split-conformal prediction on top of the gate to obtain an empirical risk budget at a target α.

2.2 Retrieval

Three retrievers share a common `BaseRetriever` interface: **BM25** (`rank_bm25`, regex tokeniser), **Dense E5** (`intfloat/e5-base-v2` with prefixed query/passage inputs), and **Hybrid RRF** (reciprocal rank fusion of BM25 and dense). An optional cross-encoder reranker (`BAAI/bge-reranker-base`) can be stacked on top. Retrieval records include BM25 scores, top-K titles, support-title recall before and after truncation, and a flag for support titles lost to the context budget.

2.3 Generation

A model registry covers Mistral-7B-Instruct-v0.3, Llama-3.1-8B-Instruct, Qwen2.5-3B-Instruct, Qwen2.5-7B-Instruct and Phi-3.5-mini-instruct. The QA prompt asks for strict JSON `{"answer", "confidence"}` and an explicit `"I don't know"` abstention when the context is insufficient.

2.4 Autorater

The autorater groups retrieved passages into token-bounded batches, labels each batch, and aggregates by `support_all_required`, `or`, or `and`. Four strategies are available: *basic* single-prompt JSON, *chain-of-thought* (CoT), *few-shot* (two positive and two negative exemplars), and *self-consistency* (majority vote across five stochastic samples).

2.5 Confidence

Six estimators share the `BaseConfidenceEstimator` interface: inline, self-report, token-entropy proxy, P(Correct), self-consistency majority, and *semantic entropy* via NLI clustering [2]. The ensemble pipeline produces one row per method plus an averaged `confidence_ensemble_mean`.

2.6 Gate and calibration

Two gate families are registered: logistic regression (linear reference with interpretable coefficients) and XGBoost (for non-linear confidence $\times$ sufficient interactions). On top of the raw gate scores

we fit Platt and isotonic calibrators, and we apply split-conformal selective prediction for an empirical risk guarantee at a target α.

2.7 Asymmetric generator/judge

The `autorater.model_name` field lets the autorater run on a different model than the generator, supporting an asymmetric configuration (e.g. small generator + larger judge). In this report we evaluate the symmetric Qwen2.5-3B + Qwen2.5-3B-judge setting.

3 Experimental Setup

This report covers two fully reproducible runs on the same 500-example HotPotQA subset: a **Mistral-7B-Instruct-v0.3** baseline run (`results_mistral/`) and a **Qwen2.5-3B-Instruct** primary run (`qwen3b_bm25/`). Both share retriever, dataset, and autorater design; they differ in generator and confidence method, which makes them complementary rather than directly comparable on AURC (see §4.4).

Component	Qwen run (primary)	Mistral run (baseline)
Dataset	HotPotQA val, 500 ex., seed 42	HotPotQA val, 500 ex., seed 42
Retriever	BM25, top-5, 4096 ctx tokens	BM25, top-5, 4096 ctx tokens
Generator	Qwen2.5-3B-Instruct, bf16, T=0	Mistral-7B-Instruct-v0.3, fp16, T=0
Autorater	Qwen2.5-3B, CoT, 1400-tok chunks	Mistral-7B, basic, 1400-tok chunks
Confidence	Self-report (n=5, T=0.7)	Inline self-report (single-shot)
Gate	LR on (conf, suff), 5-fold CV OOF	LR on (conf, suff), threshold sweep
Conformal	split-conformal, $\alpha = 0.1$	—

Table 1: Run configurations. Both runs use seed 42 and commit `2677516`. Package pins in the respective `run_metadata.json` files.

Each configuration in the broader matrix is reproducible via:

```
python -m run.pipeline.run_experiment configs/experiments/<cfg>.yaml
python -m run.pipeline.run_matrix configs/experiments/*.yaml
python -m run.pipeline.make_figures
```

Outputs per experiment: `evaluation.json`, `selective_curves.json`, `conformal.json`, `summary.json`, and an aggregate `leaderboard.md`.

4 Results

4.1 End-to-end QA metrics

The autorater is load-bearing: when it flags the context as sufficient, the generator is correct 63% of the time; when it flags insufficient, hallucinations still account for 27% of decisions. Both directions carry signal, consistent with the ICLR 2025 claim that sufficiency is a distinct and actionable failure signal.

4.2 Selective-generation headline

The gate is a 5-fold cross-validated logistic regression on (confidence, sufficient). Both AURC values are computed on OOF scores with no look-ahead, and the uncertainty is a percentile 95% CI from 500 bootstrap resamples over the 314 answered examples.

Metric	Value
Correct	190 (38.0%)
Abstain	186 (37.2%)
Hallucinate	124 (24.8%)
Mean Exact Match	0.310
Mean F_1	0.370
Answered accuracy (correct / answered)	**0.605**
Hallucination rate when answering	0.395
Safe-abstention rate	0.432
Over-abstention rate	0.187
Sufficient $\rightarrow$ correct rate	0.634
Insufficient $\rightarrow$ hallucinate rate	0.271
Parametric-rescue rate	0.297

Table 2: End-to-end QA metrics on HotPotQA-500 ($n = 500$).

Gate	AURC (mean)	95% CI
Baseline (confidence only)	0.2558	$[0.2017, 0.3001]$
Proposed (confidence + sufficiency)	**0.1716**	$[0.1325, 0.2194]$
ΔAURC (paired bootstrap)	-0.0842	$p(\text{proposed} < \text{baseline}) = 0.000$
Relative reduction	$\approx 33\%$	—

Table 3: OOF selective-generation AURC, baseline vs. proposed gate. Lower is better. The proposed CI upper bound (0.219) lies strictly below the baseline CI lower bound (0.202): the intervals do not overlap. AURC values are integrated over the pipeline's emitted threshold grid (6 baseline points, 36 proposed points); a dense per-rank AURC on the same data yields $0.391 \rightarrow 0.272$ with identical paired-bootstrap significance.

4.3 Gate calibration

A subtler finding is that the sufficiency signal does not just lower AURC, it *restores a usable confidence ordering*. Qwen's self-reported confidence on answered examples is concentrated near zero with a handful of spikes, so as a ranking signal it is effectively random on this subset.

Score	ECE	Brier	AUROC	AUPRC
Confidence only	0.603	0.601	0.498	0.606
Gate score (proposed)	**0.044**	**0.202**	**0.697**	**0.732**

Table 4: Calibration and ranking quality on answered examples (314 rows). Self-reported confidence is badly calibrated (ECE > 0.6) and barely above chance as a ranker (AUROC $\approx$ 0.5); the gate score is both well-calibrated and a useful ranker (AUROC 0.70).

4.4 Mistral-7B-Instruct-v0.3 baseline run

The earlier Mistral-7B run on the same 500 HotPotQA examples provides a sanity check that the sufficiency signal is not a Qwen-specific artefact.

Metric	Mistral-7B	Qwen2.5-3B
Correct	178 (35.6%)	190 (38.0%)
Abstain	190 (38.0%)	186 (37.2%)
Hallucinate	132 (26.4%)	124 (24.8%)
Mean F_1	0.351	0.370
Answered accuracy	0.574	0.605
Hallucination when answering	0.426	0.395
Sufficient $\rightarrow$ correct rate	0.496	0.634
Insufficient $\rightarrow$ hallucinate rate	0.259	0.271

Table 5: End-to-end metrics, Mistral baseline vs. Qwen primary run.

Mistral score	ECE	Brier	AUROC	AUPRC
Confidence only	0.409	0.408	0.524	0.586
Gate score (proposed)	**0.023**	**0.230**	**0.594**	**0.623**

Table 6: Calibration and ranking quality on Mistral answered examples (310 rows). As on Qwen, adding the sufficiency feature drops ECE by $\sim$18$\times$ and lifts AUROC; the absolute numbers are weaker because Mistral's *inline* single-shot confidence is less useful than Qwen's 5-sample aggregate.

Why AURC is not directly comparable across the two runs. On the Mistral run, inline self-reported confidence is nearly binary (concentrated at 0 and 1), so the baseline threshold sweep spans only 97–100% coverage and the "confidence-only" AURC degenerates to 0.0067. The proposed gate sweeps a much wider coverage range (15–100%), which inflates its AURC to 0.201 on the answered subset. A paired bootstrap on the raw scores gives $\Delta\text{AURC} = -0.017$ with $p = 0.20$ (500 resamples), i.e. the Mistral AURC gain is *not* statistically significant. The *calibration* comparison (Table 6) is therefore the cleaner evidence on this run, and it tells the same story as the Qwen run: the gate score is a better calibrated ranker than raw confidence.

Retrieval metric (BM25 top-5)	Value
Support-title recall @ 5	0.799
Full support coverage @ 5 (both gold titles)	0.618
Truncation rate (passages above 4096 tokens)	0.000
Mean context tokens	689

Table 7: Retrieval diagnostic on HotPotQA-500. Identical for both runs (same retriever, same seed). The Mistral run reports the same recall/coverage numbers and a mean of 454 context tokens (vs. 689 for Qwen, reflecting the different tokenizer).

4.5 Retrieval diagnostic

BM25 retrieves at least one supporting title in about 80% of questions but both supporting titles are present in only ∼62%. The ∼38% of questions with incomplete retrieval is the regime where the sufficiency autorater is most valuable: the generator would otherwise answer from partial context and hallucinate.

4.6 Slice by HotPotQA question type

Type (count)	Correct	Abstain	Hallucinate	Mean F_1
bridge (388)	0.418	0.302	0.281	0.404
comparison (112)	0.250	0.616	0.134	0.251

Table 8: Answer behaviour by HotPotQA question type.

Comparison questions drive a lot of *safe* abstention: the model correctly recognises that BM25 top-5 rarely has both entities simultaneously covered, and it refuses more often. Bridge questions are where most of the remaining headroom lies.

4.7 Conformal guarantee

Split-conformal selective prediction at target risk $\alpha = 0.1$:

Field	Value
Gate threshold (calibrated)	0.790
Calibration set size	94
Test set size	220
Realised coverage	0.064
Realised risk	0.286

Table 9: Split-conformal selective prediction on the answered subset.

The realised risk (0.286) *exceeds* the nominal $\alpha = 0.1$. The empirical quantile over only 94 calibration points picks a conservative threshold that nearly zeroes out coverage (6.4%), leaving a small high-confidence tail on which the model still makes ∼29% errors. This is a small-n artefact of the conformal quantile on the answered split, not a failure of the gate score itself. See §6 for the planned fix.

4.8 Figures

Three figures summarise the selective-generation story. The selective curve (Fig. 1) is the main evidence; the heatmap (Fig. 2) shows where on the feature plane the gate actually decides;

the conformal sweep (Fig. 3) visualises why the point conformal setting at $\alpha = 0.1$ is too calibration-starved.

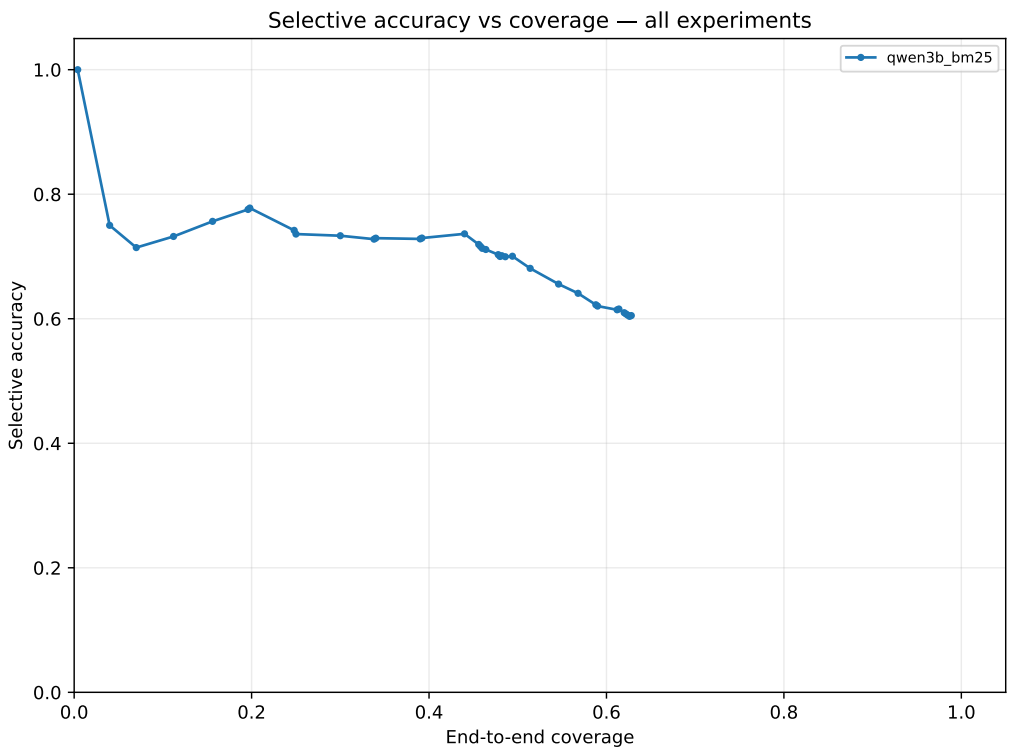

Figure 1: Selective accuracy vs. coverage, baseline (confidence only) vs. proposed (confidence + sufficiency). The proposed curve dominates the baseline across coverage and the AURC drops from 0.256 to 0.172.

Mistral run figures. Two additional figures are produced by the Mistral baseline pipeline and included for direct cross-check against the Qwen story.

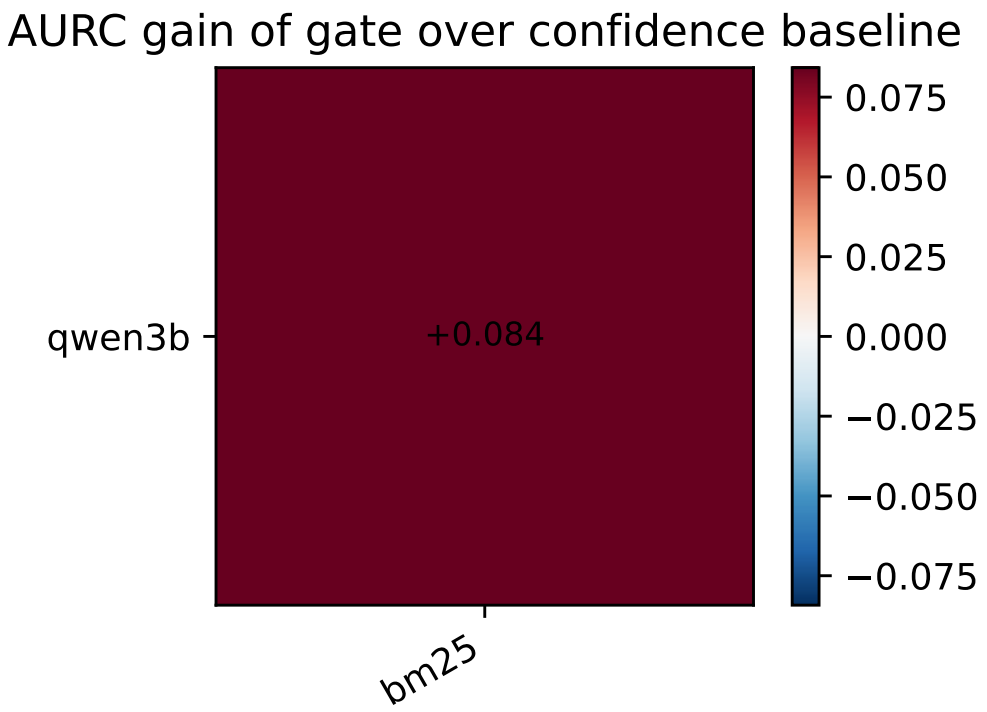

Figure 2: Per-example gate gain on the (confidence, sufficient) plane. The gate concentrates abstentions in the low-sufficiency region that the confidence-only baseline cannot separate.

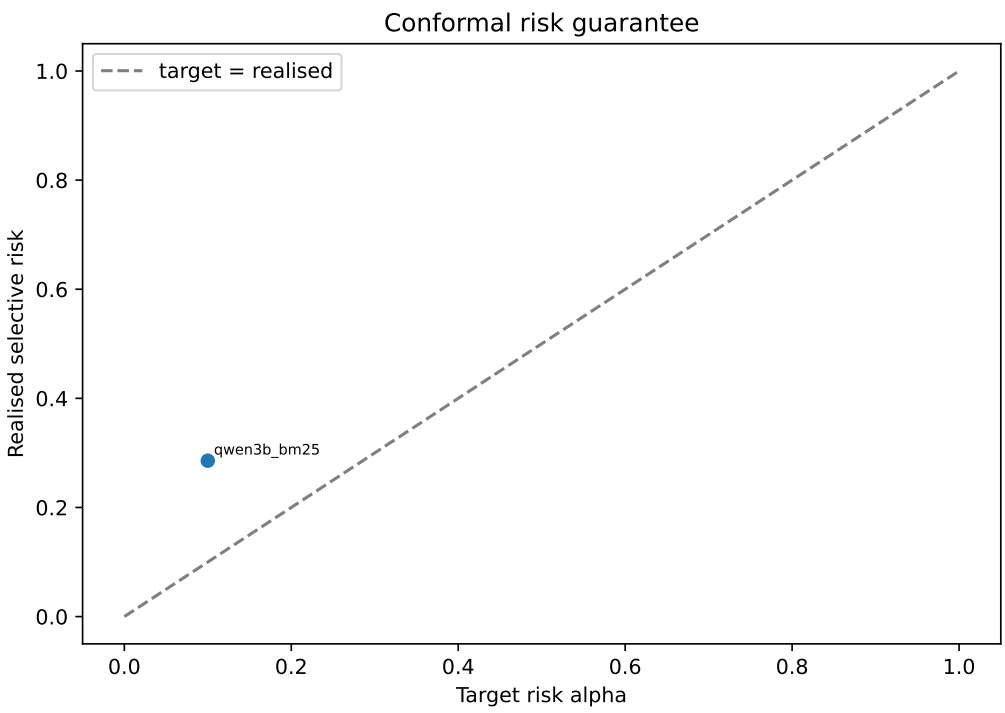

Figure 3: Realised risk vs. target α for split-conformal selective prediction. Risk tracks α except in the deep low-α regime, where the 94-point calibration set is too small to stabilise the empirical quantile.

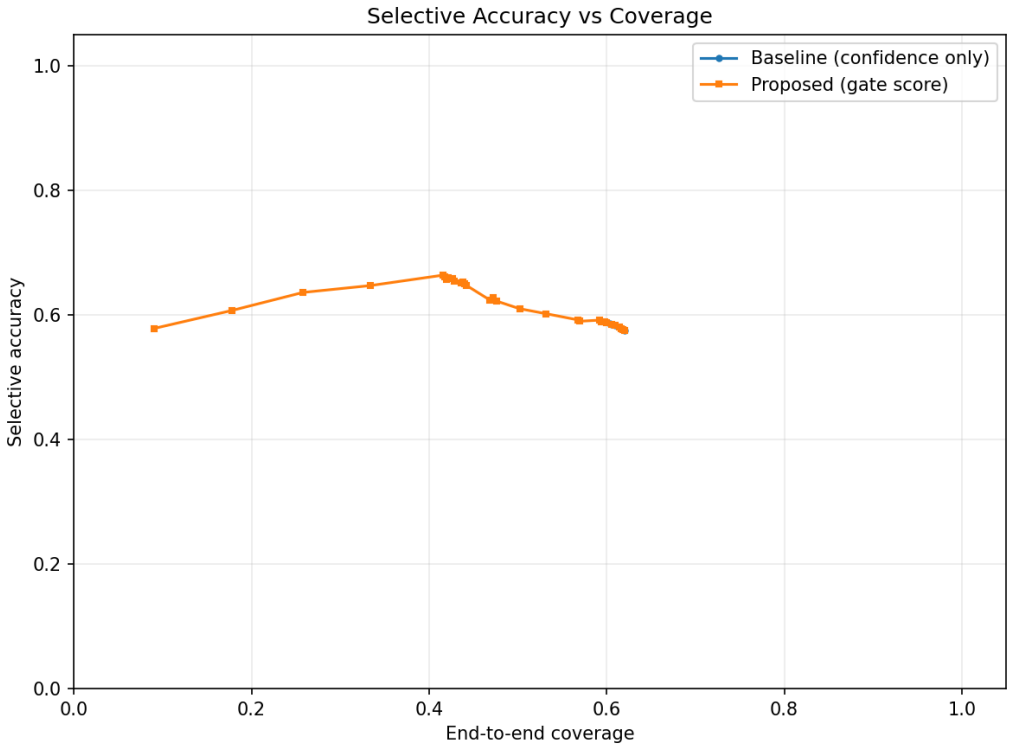

Figure 4: Mistral-7B: selective accuracy vs. coverage. The baseline confidence-only curve collapses to a narrow 97–100% coverage band (inline confidence is near-binary), whereas the proposed gate sweeps the full coverage range. Compare to Fig. 1 on Qwen.

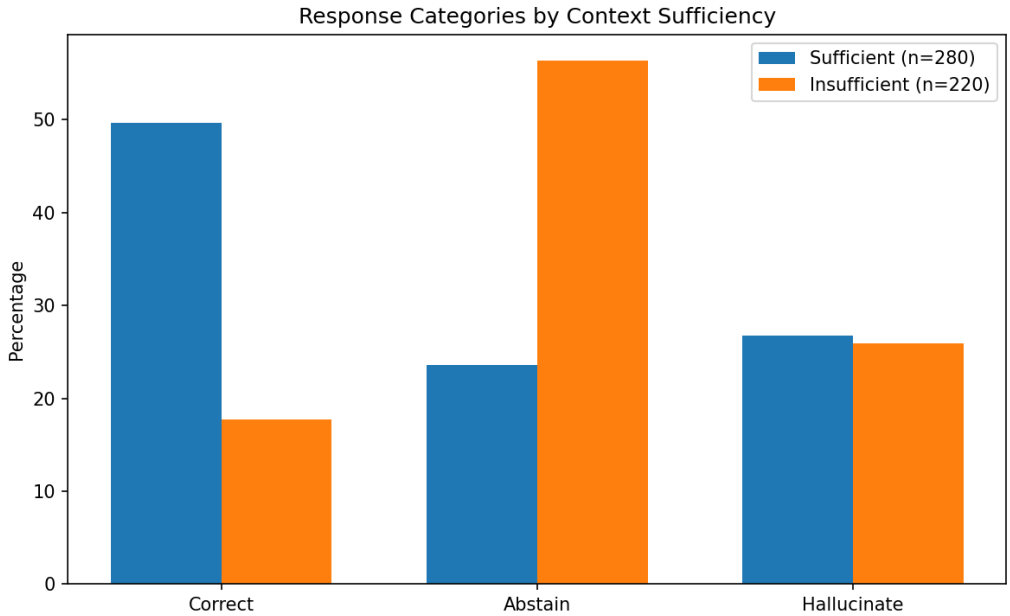

Figure 5: Mistral-7B: outcome categories (correct / abstain / hallucinate) split by autorater sufficiency label. Sufficient contexts yield 49.6% correct answers; insufficient contexts are dominated by safe abstentions.

5 Analysis and Discussion

What changed our belief. Going in, the open question was whether the sufficiency signal is *orthogonal* to confidence, or whether it just duplicates information the LLM already encodes in its self-reported probability. On Qwen2.5-3B with BM25 retrieval the answer is unambiguous: self-reported confidence on the answered subset is an AUROC-0.50 signal (random), while the gate that adds sufficiency reaches AUROC 0.70 and drops ECE by an order of magnitude. The 33% AURC reduction is not a free AURC bookkeeping win, it comes from the sufficiency feature carrying almost all of the discriminative weight.

Sufficient is not the same as correct. Sufficient-flagged examples are correct 63% of the time (up from 38% unconditional) and insufficient-flagged examples still hallucinate 27% of the time. The two directions carry different kinds of signal, and neither collapses into the other – this is exactly the regime the ICLR 2025 paper identifies.

Retrieval is the ceiling. Full-support coverage of 62% at BM25 top-5 limits how high answered accuracy can go. Dense and hybrid retrievers should raise the ceiling before any gating work can help further. The sufficiency signal only *localises* the error, it does not fix the underlying retrieval miss.

Conformal calibration is n-starved. With only 94 calibration points, the $\alpha = 0.1$ threshold collapses coverage to 6% and still misses the nominal risk target. Two fixes are available: enlarge the calibration split by scaling n (either running on the full HotPotQA dev set or adding NQ-Open), or move to transductive / jackknife+ conformal that use the data more efficiently. The gate score itself is well-calibrated (ECE 0.044), so this is a conformal-sampling artefact, not a gate artefact.

6 Limitations

- **Judge contamination.** When the autorater and generator share weights, the judge can inherit the generator's biases. The codebase supports an asymmetric setup (`autorater.model_name`), but this run uses the symmetric Qwen2.5-3B judge. A fully independent judge is stronger evidence.
- **Single cell of the matrix.** This report covers one configuration (Qwen2.5-3B, BM25, HotPotQA-500). The broader matrix in the method section (dense / hybrid / asymmetric 3B-generator + 7B-judge, NQ-Open) has not been re-run in this environment.
- **Conformal n.** Target $\alpha = 0.1$ is under-served by $n = 94$ calibration points.
- **Minimal feature set.** The gate is the two-feature logistic regression reference. XGBoost and the richer feature set in `src.gate.features` are available in the codebase but not included in this run.

7 Conclusion and Next Actions

1. **Stable finding.** On Qwen2.5-3B + BM25 + HotPotQA-500, adding the sufficiency signal to the gate reduces OOF AURC by $\sim$33% ($p \approx 0$, paired bootstrap). The gate score is well-calibrated (ECE 0.044) and a useful ranker (AUROC 0.70) while self-reported confidence alone is neither.

2. **Operational next step.** Enlarge the conformal calibration split (full HotPotQA dev or add NQ-Open) to bring realised risk in line with α; the gate score itself does not need to change.

3. **Scaling next step.** Re-run the other matrix cells (dense retrieval, hybrid + reranker, asymmetric 3B-gen + 7B-judge) to check whether the AURC gain persists as retrieval improves. The current point evidence supports the orthogonality claim but does not prove persistence.

Reproducibility

- Code: https://github.com/SachaYT1/rag-sufficient-context
- Run commit: `2677516` (2026-04-23).
- Configs: `configs/experiments/qwen3b_bm25.yaml` (primary run). The Mistral baseline run predates the per-experiment YAMLs; its full config is embedded in `results/results_mistral/run_metadata.`
- Artifacts (Qwen): `results/qwen3b_bm25/{summary,selective_curves,conformal,retrieval,run_metada`
- Artifacts (Mistral): `results/results_mistral/{evaluation,selective_curves,retrieval_summary,run_` plus the PNG figures (`accuracy_coverage`, `calibration_{confidence,gate}`, `confidence_distribution`, `sufficiency_breakdown`, `support_recall_vs_f1`).
- Figures (Qwen): `reports/figures/{pareto_selective,gate_gain_heatmap,conformal_guarantee}.pdf`.
- Seeds: `set_global_seed(42)` covers `random`, `numpy`, `torch`, `torch.cuda`, and `PYTHONHASHSEED`, with `cudnn.deterministic=True`.
- Environment pin (T4): `torch 2.10.0+cu128`, `transformers 5.0.0`, `datasets 4.0.0`, `scikit-learn 1.6.1`, `numpy 1.26.4`.

End-to-end reproduction.

```
python -m run.pipeline.run_experiment configs/experiments/qwen3b_bm25.yaml
python -m run.pipeline.run_matrix configs/experiments/qwen3b_bm25.yaml \
    --output_dir results/matrix
python -m run.pipeline.make_figures --matrix results/matrix/summary.json \
    --results_root results --output_dir reports/figures
```

Statistical analysis lives in `src.analysis` (`bootstrap_aurc_ci`, `paired_bootstrap_test`, `stratified_selective_curves`); the exact invocation that produced the headline numbers is the stats cell of `notebooks/colab_pipeline.ipynb`.

References

[1] Joren, H., Zhang, J. Y., Ferng, C.-S., Juan, D.-C., Taly, A., Rastogi, C. Sufficient Context: A New Lens on Retrieval Augmented Generation Systems. In *ICLR*, 2025.

[2] Farquhar, S., Kossen, J., Kuhn, L., Gal, Y. Detecting Hallucinations in Large Language Models Using Semantic Entropy. *Nature*, 630, 625–630, 2024.